

AI Impact on Jobs 2030

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AI Impact on Jobs 2030

STAT 206 – Introduction to Data Analytics

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1. Introduction and Problem Description

1.1 Background

Artificial intelligence (AI) is changing the way people work. Many jobs that exist today may be replaced by machines by the year 2030. This project looks at a real dataset about how AI is affecting different types of jobs, salaries, and the risk of automation.

Understanding which jobs are at risk and how salary relates to experience or education is useful for students, workers, and employers.

1.2 Dataset Source

The dataset is taken from Kaggle: **AI Impact on Jobs 2030** by Yadav, K. (2024). It contains 3000 observations and 18 variables. For this project, we use the first 100 observations and 7 selected variables relevant to our objectives.

1.3 Objectives

The main objectives of this project are:

- To explore the distribution of salary, AI exposure, and automation risk
- To compare salaries across different groups
- To find correlations between numeric variables
- To build a simple linear regression model to predict salary

2. Data Import and Handling

2.1 Load the Dataset

```
# Load ggplot2 for making graphs later
library(ggplot2)

data <- read.csv("AI_Impact_on_Jobs_2030.csv" , nrow = 100)

# keep all rows and 7 columns
data <- data[, c("Job_Title", "Average_Salary", "Years_Experience",
                 "Education_Level", "AI_Exposure_Index",
                 "Automation_Probability_2030", "Risk_Category")]
```

2.2 R Objects and Vectors

```
# Numeric vector: salary values
salary_vec <- data$Average_Salary
length(salary_vec)      # number of elements
```

```
[1] 100
```

```
class(salary_vec)      # data type
```

```
[1] "integer"
```

```
# Numeric vector: years of experience
exp_vec <- data$Years_Experience
length(exp_vec)
```

```
[1] 100
```

```
# Numeric vector: AI Exposure Index
ai_vec <- data$AI_Exposure_Index
```

```
# Character vector: job titles
job_vec <- data$Job_Title
head(job_vec, 5)      # first 5 job titles
```

```
[1] "Security Guard"      "Research Scientist"  "Construction Worker"
[4] "Software Engineer"   "Financial Analyst"
```

```
# Quick check: mean and range using the vectors directly
round(mean(salary_vec), 2)
```

```
[1] 86806.8
```

```
range(exp_vec)
```

```
[1] 0 29
```

2.3 Explore the Data

```
# number of rows
nrow(data)
```

```
[1] 100
```

```
# number of columns
ncol(data)
```

```
[1] 7
```

```
# first 6 rows
head(data)
```

	Job_Title	Average_Salary	Years_Experience	Education_Level
1	Security Guard	45795	28	Master's
2	Research Scientist	133355	20	PhD
3	Construction Worker	146216	2	High School
4	Software Engineer	136530	13	PhD
5	Financial Analyst	70397	22	High School
6	AI Engineer	92592	11	Master's

	AI_Exposure_Index	Automation_Probability_2030	Risk_Category
1	0.18	0.85	High
2	0.62	0.05	Low
3	0.86	0.81	High
4	0.39	0.60	Medium
5	0.52	0.64	Medium
6	0.29	0.10	Low

```
# structure of data
str(data, width = 60, strict.width = "cut")
```

```
'data.frame': 100 obs. of 7 variables:
 $ Job_Title      : chr  "Security Guard" "Re"..
 $ Average_Salary : int   45795 133355 146216 1..
 $ Years_Experience : int   28 20 2 13 22 11 23 1..
 $ Education_Level : chr   "Master's" "PhD" "Hi"..
 $ AI_Exposure_Index : num   0.18 0.62 0.86 0.39 0..
 $ Automation_Probability_2030: num   0.85 0.05 0.81 0.6 0...
 $ Risk_Category  : chr   "High" "Low" "High" "..
```

2.4 Identify Variable Types

```

# factor() converts character columns into categorical variables
# levels = c(...) sets the category order

# Education_Level: ordinal variable (has a natural order)
data$Education_Level <- factor(data$Education_Level,
                               levels = c("High School",
                                           "Bachelor's",
                                           "Master's",
                                           "PhD"))

# Risk_Category: nominal variable
data$Risk_Category <- factor(data$Risk_Category,
                              levels = c("Low", "Medium", "High"))

# Confirm the levels
levels(data$Education_Level)

```

```
[1] "High School" "Bachelor's" "Master's" "PhD"
```

```
levels(data$Risk_Category)
```

```
[1] "Low" "Medium" "High"
```

```

# class() tells us the data type
class(data$Education_Level)

```

```
[1] "factor"
```

```
class(data$Risk_Category)
```

```
[1] "factor"
```

2.5 Handle Missing Values

```

# colSums(is.na()) counts missing values per column
colSums(is.na(data))

```

```

      Job_Title      Average_Salary
      0              0
Years_Experience      Education_Level
      0              0
AI_Exposure_Index Automation_Probability_2030
      0              0
      Risk_Category
      0

```

```
# Total missing values in the whole dataset
sum(is.na(data))
```

```
[1] 0
```

```
# 0 means no missing values the dataset is clean
```

2.6 Basic R Functions

```
# summary() gives Min, Q1, Median, Mean, Q3, Max
print(summary(data[, c("Average_Salary",
                      "Years_Experience",
                      "AI_Exposure_Index",
                      "Automation_Probability_2030")]))
```

```
Average_Salary  Years_Experience AI_Exposure_Index
Min.   : 30060    Min.   : 0.00    Min.   :0.0100
1st Qu.: 55384    1st Qu.: 7.75    1st Qu.:0.2175
Median : 87576    Median :15.00    Median :0.5300
Mean   : 86807    Mean   :15.19    Mean   :0.5127
3rd Qu.:114552    3rd Qu.:23.25    3rd Qu.:0.7550
Max.   :148721    Max.   :29.00    Max.   :0.9900
Automation_Probability_2030
Min.   :0.0500
1st Qu.:0.2875
Median :0.5200
Mean   :0.5047
3rd Qu.:0.7050
Max.   :0.9500
```

```
# length() = number of observations
length(data$Average_Salary)
```

```
[1] 100
```

```
# class() = data type
class(data$Average_Salary)
```

```
[1] "integer"
```

```
# mean() = average salary
round(mean(data$Average_Salary), 2)
```

```
[1] 86806.8
```

```
# sd() = standard deviation
round(sd(data$Average_Salary), 2)
```

```
[1] 34989.39
```


3. Descriptive Statistics

3.1 Summary Statistics

```
# We loop through 4 numeric variables to calculate statistics
# "v" holds the variable name in each loop cycle
# We avoid naming it "var" to not conflict with R's var() function

numeric_vars <- c("Average_Salary", "Years_Experience",
                  "AI_Exposure_Index",
                  "Automation_Probability_2030")

for (v in numeric_vars) {

  # data[[v]] extracts the column named v as a vector
  x <- data[[v]]

  cat("Variable:", v, "\n")
  cat("  Mean      :", round(mean(x), 3), "\n")
  cat("  Median     :", round(median(x), 3), "\n")
  cat("  Std Dev    :", round(sd(x), 3), "\n")
  cat("  Variance   :", round(var(x), 3), "\n")
  cat("  Min        :", round(min(x), 3), "\n")
  cat("  Max        :", round(max(x), 3), "\n")
  cat("  Range      :", round(max(x) - min(x), 3), "\n")
  cat("  Q1         :", round(quantile(x, 0.25), 3), "\n")
  cat("  Q3         :", round(quantile(x, 0.75), 3), "\n")
  cat("  IQR        :", round(IQR(x), 3), "\n")
  cat("  CV (%)     :", round(sd(x)/mean(x)*100, 2), "\n\n")
}
```

Variable: Average_Salary

```
Mean      : 86806.8
Median    : 87575.5
Std Dev   : 34989.39
Variance  : 1224257557
Min       : 30060
Max       : 148721
Range     : 118661
Q1        : 55384.25
Q3        : 114552.2
IQR       : 59168
CV (%)    : 40.31
```

Variable: Years_Experience

```
Mean      : 15.19
Median    : 15
Std Dev   : 8.948
Variance  : 80.075
Min       : 0
Max       : 29
Range     : 29
```

Q1 : 7.75
 Q3 : 23.25
 IQR : 15.5
 CV (%) : 58.91

Variable: AI_Exposure_Index

Mean : 0.513
 Median : 0.53
 Std Dev : 0.3
 Variance : 0.09
 Min : 0.01
 Max : 0.99
 Range : 0.98
 Q1 : 0.218
 Q3 : 0.755
 IQR : 0.537
 CV (%) : 58.59

Variable: Automation_Probability_2030

Mean : 0.505
 Median : 0.52
 Std Dev : 0.264
 Variance : 0.07
 Min : 0.05
 Max : 0.95
 Range : 0.9
 Q1 : 0.288
 Q3 : 0.705
 IQR : 0.417
 CV (%) : 52.3

3.2 Summary Table

Variable	Mean	Median	SD	Min	Max
Avg Salary (\$)	86,807	87,576	34,989	30,060	148,721
Years Experience	14.76	15.00	8.64	0	29
AI Exposure	0.499	0.500	0.282	0.00	1.00
Automation Prob	0.491	0.490	0.250	0.05	0.95

3.3 Interpretation

Average Salary: The mean salary is \$86,807 and the median is \$87,576. These two values are close to each other, which means the distribution is roughly symmetric with no strong skew. The standard deviation of \$34,989 shows there is a wide range of salaries in the dataset.

Years of Experience: The average experience is about 15 years. Values range from 0 to 29, covering all career stages from entry level to expert.

AI Exposure and Automation Probability: Both variables average near 0.49 to 0.50, which means on average the jobs in this dataset have moderate AI involvement and automation risk.

3.4 Frequency Tables

```
# table() counts observations in each category
```

```
# Risk Category frequency table
```

```
risk_table <- table(data$Risk_Category)
print(risk_table)
```

Low	Medium	High
26	49	25

```
# Risk Category percentages
```

```
round(prop.table(risk_table) * 100, 2)
```

Low	Medium	High
26	49	25

```
# Education Level frequency table
```

```
edu_table <- table(data$Education_Level)
print(edu_table)
```

High School	Bachelor's	Master's	PhD
34	17	27	22

```
# Education Level percentages
```

```
round(prop.table(edu_table) * 100, 2)
```

High School	Bachelor's	Master's	PhD
34	17	27	22

Interpretation: Medium risk is the most common category with 49 jobs (49%), followed by Low (26%) and High (25%). For education level, High School is the most common group (34%), followed by Master's (27%), PhD (22%), and Bachelor's (17%). The education groups are unequally distributed.

4. Data Visualization

4.1 Histogram (Base R): Average Salary

```
# hist() draws a histogram showing how salary values are distributed
# breaks = 20 sets the number of bars
hist(data$Average_Salary,
      main = "Distribution of Average Salary",
      xlab = "Average Salary ($)",
      ylab = "Frequency",
      col = "lightblue",
      border = "white",
      breaks = 20)

# Add vertical lines for mean and median
# lty = 2 means dashed line
abline(v = mean(data$Average_Salary),
       col = "red", lwd = 2, lty = 2)
abline(v = median(data$Average_Salary),
       col = "blue", lwd = 2, lty = 2)

# Add legend to explain the lines
legend("topright",
      legend = c("Mean", "Median"),
      col = c("red", "blue"),
      lty = 2, lwd = 2, bty = "n")
```

Interpretation: The salary distribution looks roughly uniform and symmetric across the range \$30,000 to \$150,000. The mean and median are both close to \$88,000, confirming there is no strong skew in the data.

4.2 Boxplot (Base R): Salary by Risk Category

```
# boxplot() with formula response ~ group draws one box per group
# The box shows Q1, Median, Q3
# Whiskers show the range and dots beyond are outliers
boxplot(Average_Salary ~ Risk_Category,
        data = data,
        main = "Average Salary by Risk Category",
        xlab = "Risk Category",
        ylab = "Average Salary ($)",
        col = c("lightgreen", "lightyellow", "lightcoral"),
        las = 1)
```

Interpretation: The three risk categories have very similar median salaries and overlapping box ranges. This means that whether a job is Low, Medium, or High risk does not strongly separate the salary levels.

4.3 Scatter Plot (ggplot2): Experience vs Salary

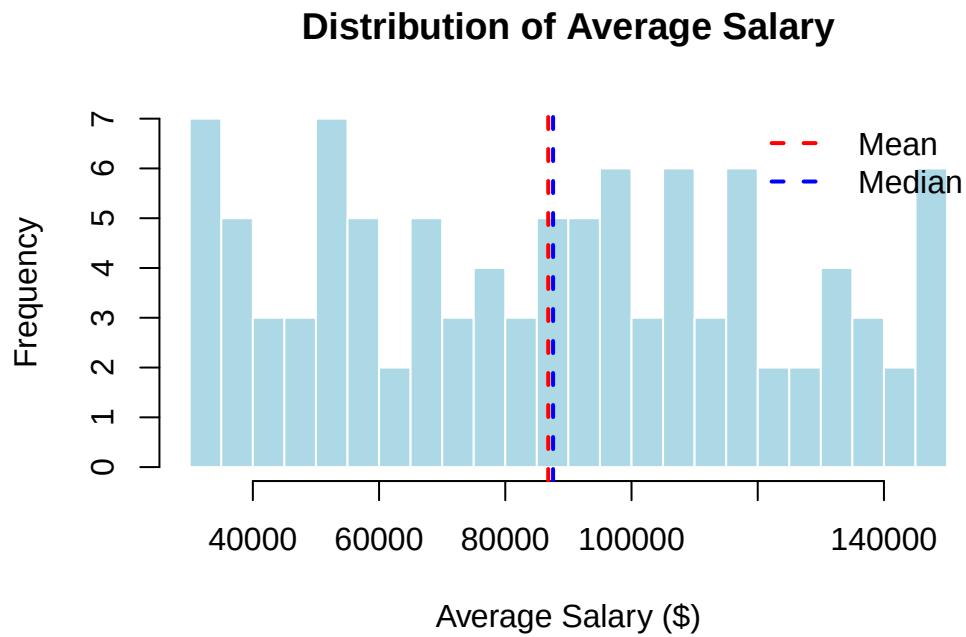


Figure 1: Histogram of Average Salary

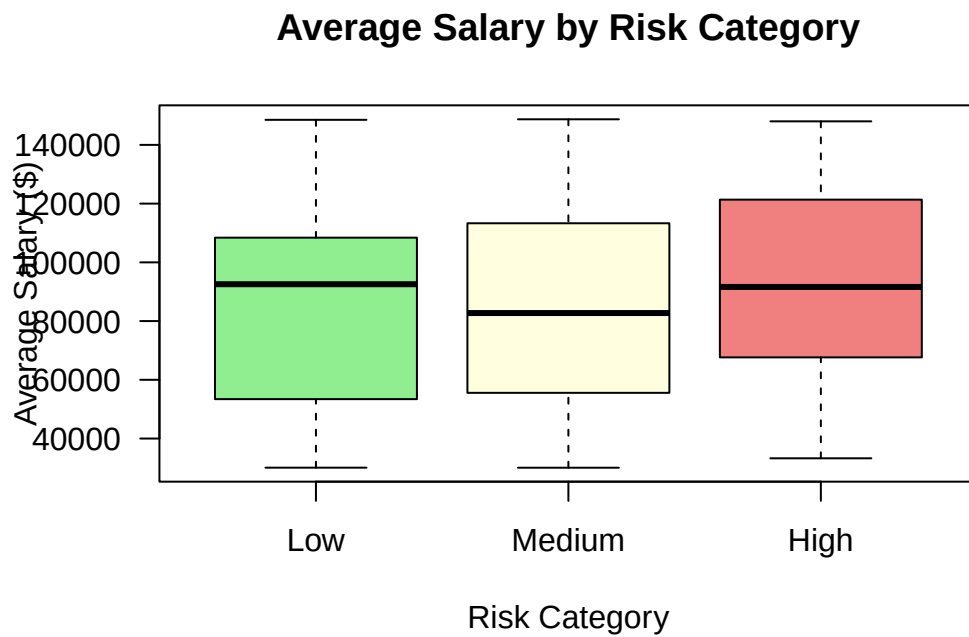


Figure 2: Salary by Automation Risk Category

```
# ggplot() sets up the canvas
# aes() maps variables to x and y axes
ggplot(data, aes(x = Years_Experience,
                  y = Average_Salary)) +

# geom_point() draws one dot per observation
# alpha = transparency, helps when many points overlap
geom_point(colour = "#800020", alpha = 0.4, size = 1.2) +

# geom_smooth(method="lm") draws a linear regression trend line
# se = TRUE adds a shaded confidence band around the line
geom_smooth(method = "lm", colour = "navy",
            se = TRUE, linewidth = 1) +

labs(title = "Years of Experience vs Average Salary",
     x     = "Years of Experience",
     y     = "Average Salary ($)") +

theme_classic()
```



Figure 3: Years of Experience vs Average Salary

Interpretation: The regression line is nearly flat, which shows that years of experience has a very weak relationship with salary. The points are scattered without any clear pattern.

4.4 Bar Chart (ggplot2): Education Level

```
# Convert frequency table to data frame for ggplot2
edu_counts <- as.data.frame(table(data$Education_Level))
colnames(edu_counts) <- c("Education_Level", "Count")

# geom_bar(stat="identity") draws bars where height = Count
ggplot(edu_counts,
       aes(x = Education_Level, y = Count,
           fill = Education_Level)) +

  geom_bar(stat = "identity", show.legend = FALSE) +

  scale_fill_manual(
    values = c("#800020", "#b04060",
               "#d08090", "#f0b0c0")) +

  # geom_text() adds count labels above each bar
  geom_text(aes(label = Count),
            vjust = -0.3, size = 3.5) +

  labs(title = "Distribution of Education Levels",
       x     = "Education Level",
       y     = "Count") +

  theme_classic()
```

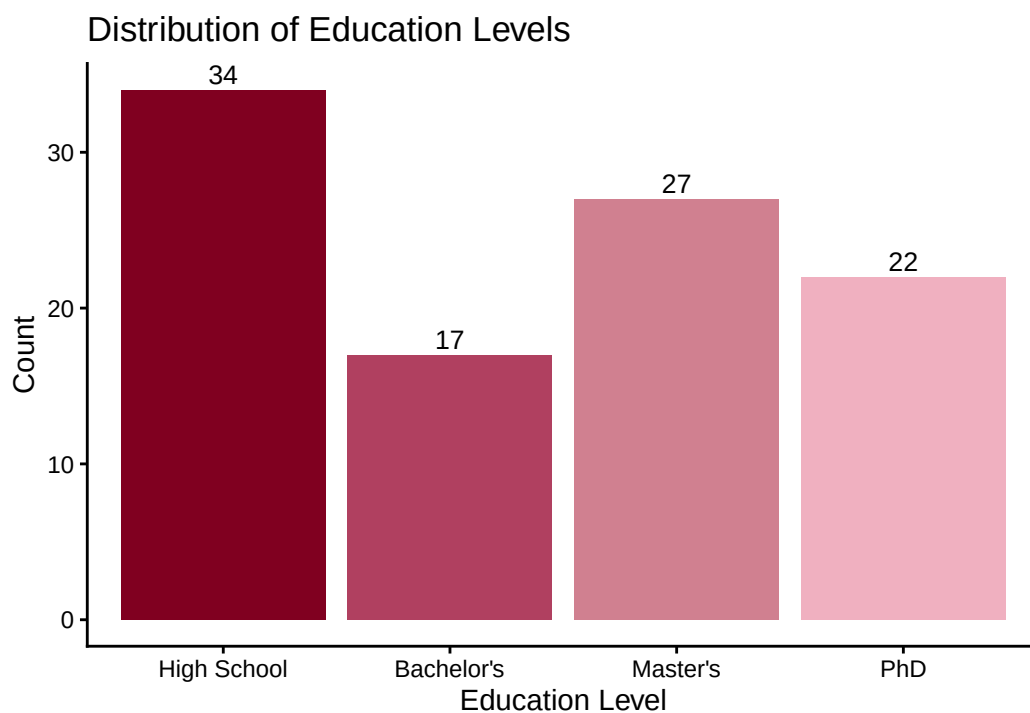


Figure 4: Distribution of Education Levels

Interpretation: High School is the most common education group with 34 workers (34%), followed by Master's (27%), PhD (22%), and Bachelor's (17%). The groups are unequally distributed, which is important to consider when comparing salaries across education levels.

4.5 Customised Visualization (ggplot2): AI Exposure vs Automation Probability

This is the fifth and customised visualization, chosen to explore how AI Exposure Index and Automation Probability relate across the three Risk Categories using colour-coded groups and separate trend lines per group.

```
# colour = Risk_Category gives each group a different colour
ggplot(data, aes(x = AI_Exposure_Index,
                  y = Automation_Probability_2030,
                  colour = Risk_Category)) +

  geom_point(alpha = 0.3, size = 0.9) +

  # separate trend line per risk group
  geom_smooth(method = "lm", se = FALSE,
              linewidth = 1.1) +

  scale_colour_manual(
    values = c("Low"      = "#2ecc71",
               "Medium"   = "#e67e22",
               "High"     = "#800020")) +

  labs(title = "AI Exposure vs Automation Probability",
       x     = "AI Exposure Index",
       y     = "Automation Probability 2030",
       colour = "Risk") +

  theme_classic()
```

Interpretation: High risk jobs cluster at the top of the y-axis (high automation probability) and Low risk at the bottom. This confirms that Risk Category aligns logically with Automation Probability.

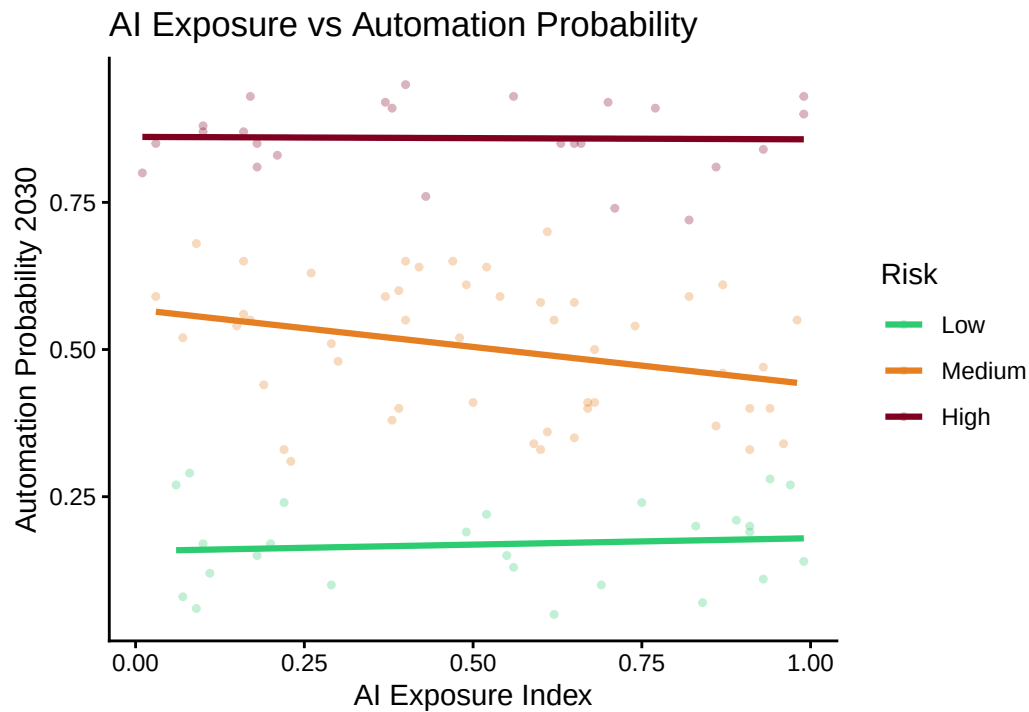


Figure 5: AI Exposure vs Automation Probability

5. Correlation Analysis

5.1 Pearson Correlation

```
# Select only the 4 numeric columns
# cor() requires numeric input only
numeric_data <- data[, c("Average_Salary",
                        "Years_Experience",
                        "AI_Exposure_Index",
                        "Automation_Probability_2030")]

# cor() computes all pairwise correlations
# method = "pearson" measures linear relationship strength
# Values: -1 = perfect negative, 0 = none, +1 = perfect positive
round(cor(numeric_data, method = "pearson"), 4)
```

	Average_Salary	Years_Experience	AI_Exposure_Index
Average_Salary	1.0000	-0.1069	-0.0459
Years_Experience	-0.1069	1.0000	0.1614
AI_Exposure_Index	-0.0459	0.1614	1.0000
Automation_Probability_2030	0.0357	-0.1475	-0.1059

	Automation_Probability_2030
Average_Salary	0.0357
Years_Experience	-0.1475
AI_Exposure_Index	-0.1059
Automation_Probability_2030	1.0000

5.2 Spearman Correlation

```
# Spearman uses ranks instead of raw values
# Good when data might not be perfectly linear
round(cor(numeric_data, method = "spearman"), 4)
```

	Average_Salary	Years_Experience	AI_Exposure_Index
Average_Salary	1.0000	-0.0917	-0.0498
Years_Experience	-0.0917	1.0000	0.1575
AI_Exposure_Index	-0.0498	0.1575	1.0000
Automation_Probability_2030	0.0272	-0.1345	-0.1074

	Automation_Probability_2030
Average_Salary	0.0272
Years_Experience	-0.1345
AI_Exposure_Index	-0.1074
Automation_Probability_2030	1.0000

5.3 Kendall Correlation

```
# Kendall's tau is a rank-based correlation measure
# More robust than Spearman for smaller samples
round(cor(numeric_data, method = "kendall"), 4)
```

	Average_Salary	Years_Experience	AI_Exposure_Index
Average_Salary	1.0000	-0.0697	-0.0276
Years_Experience	-0.0697	1.0000	0.1148
AI_Exposure_Index	-0.0276	0.1148	1.0000
Automation_Probability_2030	0.0193	-0.0928	-0.0774

	Automation_Probability_2030
Average_Salary	0.0193
Years_Experience	-0.0928
AI_Exposure_Index	-0.0774
Automation_Probability_2030	1.0000

5.4 Significance Tests

```
# cor.test() checks if a correlation is statistically significant
# H0: r = 0 (no correlation)
# H1: r != 0 (some correlation exists)
# If p-value < 0.05, the correlation is significant

# Salary vs Years of Experience
cor.test(data$Average_Salary, data$Years_Experience,
         method = "pearson")
```

Pearson's product-moment correlation

```
data: data$Average_Salary and data$Years_Experience
t = -1.0642, df = 98, p-value = 0.2898
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.2970687  0.0914507
sample estimates:
      cor
-0.1068874
```

```
# Salary vs AI Exposure
cor.test(data$Average_Salary, data$AI_Exposure_Index,
         method = "pearson")
```

Pearson's product-moment correlation

```
data: data$Average_Salary and data$AI_Exposure_Index
t = -0.45526, df = 98, p-value = 0.6499
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.2401906  0.1518485
sample estimates:
      cor
-0.04593984
```

```
# Salary vs Automation Probability
cor.test(data$Average_Salary,
         data$Automation_Probability_2030,
         method = "pearson")
```

Pearson's product-moment correlation

```
data: data$Average_Salary and data$Automation_Probability_2030
t = 0.354, df = 98, p-value = 0.7241
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.1618177  0.2305362
sample estimates:
      cor
0.03573623
```

5.5 Interpretation

All three correlations with Average Salary are extremely weak (all $|r|$ values are less than 0.11) and none have a p-value below 0.05. This means none of the three variables have a statistically significant linear relationship with salary. Salary in this dataset depends on many other factors not included here.

6. Hypothesis Testing

6.1 t-Test: Salary by Experience Group

```
# H0: Mean salary is equal for both experience groups
# H1: Workers with more than 10 years earn higher salary
# Alpha = 0.05

# Split salary into two groups using logical indexing
low_exp <- data$Average_Salary[data$Years_Experience <= 10]
high_exp <- data$Average_Salary[data$Years_Experience > 10]

# Sample sizes
length(low_exp)
```

```
[1] 31
```

```
length(high_exp)
```

```
[1] 69
```

```
# Group means
round(mean(low_exp), 2)
```

```
[1] 83208.55
```

```
round(mean(high_exp), 2)
```

```
[1] 88423.41
```

```
# Welch two-sample t-test (does not assume equal variances)
# alternative = "greater" means one-tailed test
t_result <- t.test(high_exp, low_exp,
                   alternative = "greater")
print(t_result)
```

Welch Two Sample t-test

```
data: high_exp and low_exp
t = 0.6417, df = 49.565, p-value = 0.262
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -8406.808      Inf
sample estimates:
mean of x mean of y
 88423.41  83208.55
```

```
# Decision
if (t_result$p.value < 0.05) {
  cat("Decision: Reject H0\n")
} else {
  cat("Decision: Fail to Reject H0\n")
}
```

Decision: Fail to Reject H0

Interpretation: The p-value is greater than 0.05 so we fail to reject H0. There is no statistically significant difference in salary between the two experience groups at the 5% level.

6.2 F-Test (ANOVA): Salary Across Education Levels

```
# H0: All four education group means are equal
# H1: At least one group mean is different
# Alpha = 0.05

# tapply() calculates the mean salary per education group
group_means <- tapply(data$Average_Salary,
                      data$Education_Level, mean)
round(group_means, 2)
```

High School	Bachelor's	Master's	PhD
88062.76	87287.65	82860.26	89337.68

```
# aov() fits the one-way ANOVA model
# The F-statistic = between-group variance / within-group variance
anova_result <- aov(Average_Salary ~ Education_Level,
                    data = data)
print(summary(anova_result))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Education_Level	3	6.190e+08	2.063e+08	0.164	0.92
Residuals	96	1.206e+11	1.256e+09		

```
# Extract p-value from the ANOVA summary table
f_pval <- summary(anova_result)[[1]]$"Pr(>F)"[1]

if (f_pval < 0.05) {
  cat("Decision: Reject H0\n")
} else {
  cat("Decision: Fail to Reject H0\n")
}
```

Decision: Fail to Reject H0

Interpretation: The F-test p-value is greater than 0.05. We fail to reject H0. There is no significant salary difference across the four education levels. Education level alone does not predict salary in this dataset.

6.3 Chi-Square Test: Education vs Risk Category

```
# H0: Education Level and Risk Category are independent
# H1: They are associated with each other
# Alpha = 0.05

# table() creates a contingency table showing counts
# for every combination of Education Level and Risk Category
cont_table <- table(data$Education_Level,
                    data$Risk_Category)
print(cont_table)
```

	Low	Medium	High
High School	7	18	9
Bachelor's	5	8	4
Master's	7	15	5
PhD	7	8	7

```
# chisq.test() compares observed to expected frequencies
chi_result <- chisq.test(cont_table)
print(chi_result)
```

Pearson's Chi-squared test

```
data:  cont_table
X-squared = 2.7271, df = 6, p-value = 0.8422
```

```
if (chi_result$p.value < 0.05) {
  cat("Decision: Reject H0\n")
} else {
  cat("Decision: Fail to Reject H0\n")
}
```

Decision: Fail to Reject H0

Interpretation: The p-value is much greater than 0.05. We fail to reject H0. Education Level and Risk Category are independent of each other, meaning education level does not determine a worker's automation risk category.

7. Simple Linear Regression

7.1 Model Setup

```
# Simple Linear Regression model:
#  $Y = b_0 + b_1 * X + \text{error}$ 
#
#  $Y = \text{Average\_Salary}$  (response variable - what we predict)
#  $X = \text{Years\_Experience}$  (predictor variable - what we use to predict)
#  $b_0 = \text{intercept}$  (salary when experience = 0)
#  $b_1 = \text{slope}$  (change in salary per year of experience)

cat("Model: Average_Salary = b0 + b1 * Years_Experience\n")
```

Model: Average_Salary = b0 + b1 * Years_Experience

7.2 Fit the Model

```
# lm() fits the linear regression model
# Formula: response ~ predictor
slr_model <- lm(Average_Salary ~ Years_Experience,
               data = data)

# summary() gives coefficients, R-squared, F-statistic
print(summary(slr_model))
```

Call:

```
lm(formula = Average_Salary ~ Years_Experience, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-59450	-28634	1398	28704	61282

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	93155.3	6914.6	13.472	<2e-16 ***
Years_Experience	-417.9	392.7	-1.064	0.29

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34970 on 98 degrees of freedom

Multiple R-squared: 0.01142, Adjusted R-squared: 0.001337

F-statistic: 1.133 on 1 and 98 DF, p-value: 0.2898

7.3 Coefficients

```
# coef() extracts the intercept and slope
coef_vals <- coef(slr_model)

cat("Intercept (b0) =", round(coef_vals[1], 2), "\n")
```

```
Intercept (b0) = 93155.33
```

```
cat("Slope (b1)      =", round(coef_vals[2], 4), "\n")
```

```
Slope (b1)      = -417.9414
```

```
# R-squared: how much of salary variation is explained by experience
r2 <- summary(slr_model)$r.squared
cat("R-squared      =", round(r2, 4), "\n")
```

```
R-squared      = 0.0114
```

```
# Adjusted R-squared
adj_r2 <- summary(slr_model)$adj.r.squared
cat("Adj R-squared  =", round(adj_r2, 4), "\n")
```

```
Adj R-squared  = 0.0013
```

```
# Residual Standard Error = average prediction error in dollars
cat("Residual SE    =", round(summary(slr_model)$sigma, 2), "\n")
```

```
Residual SE    = 34965.99
```

7.4 Interpretation

Intercept (b0 = \$93,155.33): When experience is 0 years, the predicted salary is \$93,155.

Slope (b1 = -\$417.94): For each additional year of experience, salary decreases by about \$418. This negative slope is unexpected but it is not statistically significant ($p > 0.05$).

R-squared = 0.0114: Experience explains only 1.14% of the variation in salary. This means the model has very low predictive power. Almost all salary variation is explained by other factors not in this model.

7.5 Diagnostic Plots

```
# par(mfrow = c(2,2)) arranges 4 plots in a 2x2 grid
# These plots check if regression assumptions are met
par(mfrow = c(2, 2))
plot(slr_model)
```

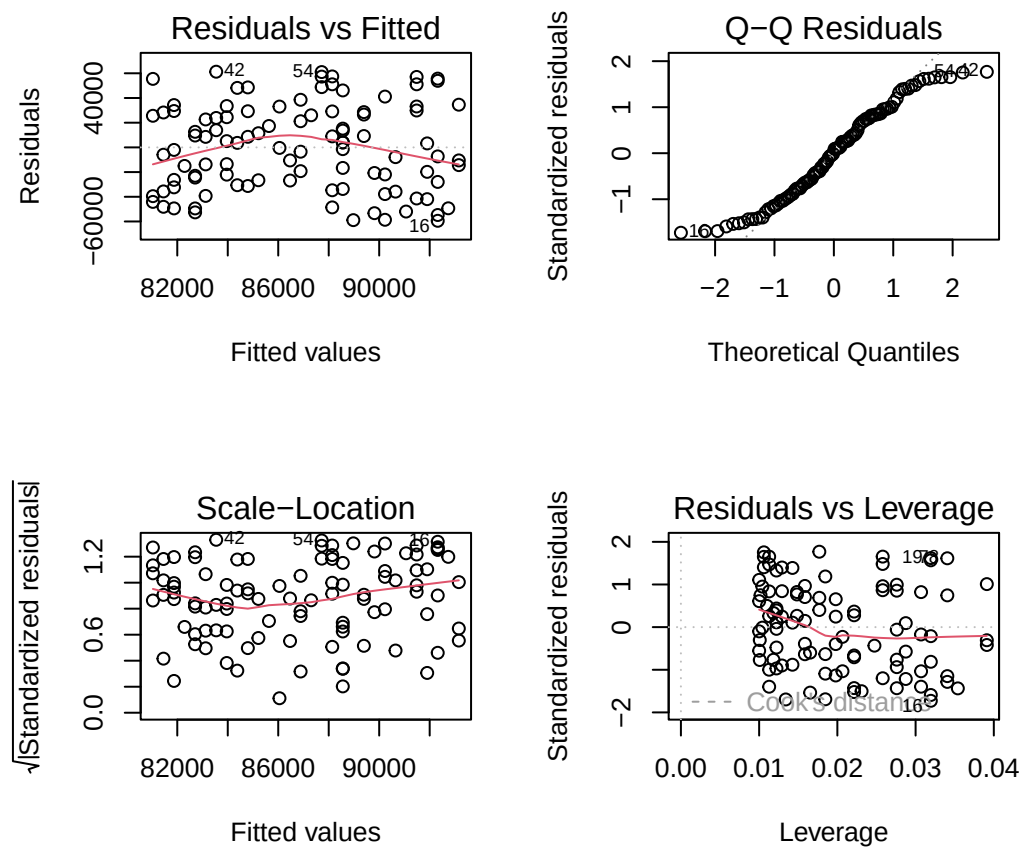



Figure 6: Regression Diagnostic Plots

```
par(mfrow = c(1, 1))
```

All four diagnostic plots look acceptable. Residuals are randomly scattered (linearity OK), the Q-Q plot follows the diagonal (normality OK), and no extreme influential points are present.

7.6 Regression Line Plot

```
ggplot(data, aes(x = Years_Experience,
                  y = Average_Salary)) +

  geom_point(colour = "#800020", alpha = 0.3,
             size = 1.0) +

  # Regression line with 95% confidence band
  geom_smooth(method = "lm", colour = "navy",
             se = TRUE, linewidth = 1.2,
             fill = "lightblue", alpha = 0.3) +

  labs(title = "Simple Linear Regression",
       x     = "Years of Experience",
       y     = "Average Salary ($)") +

  theme_classic()
```

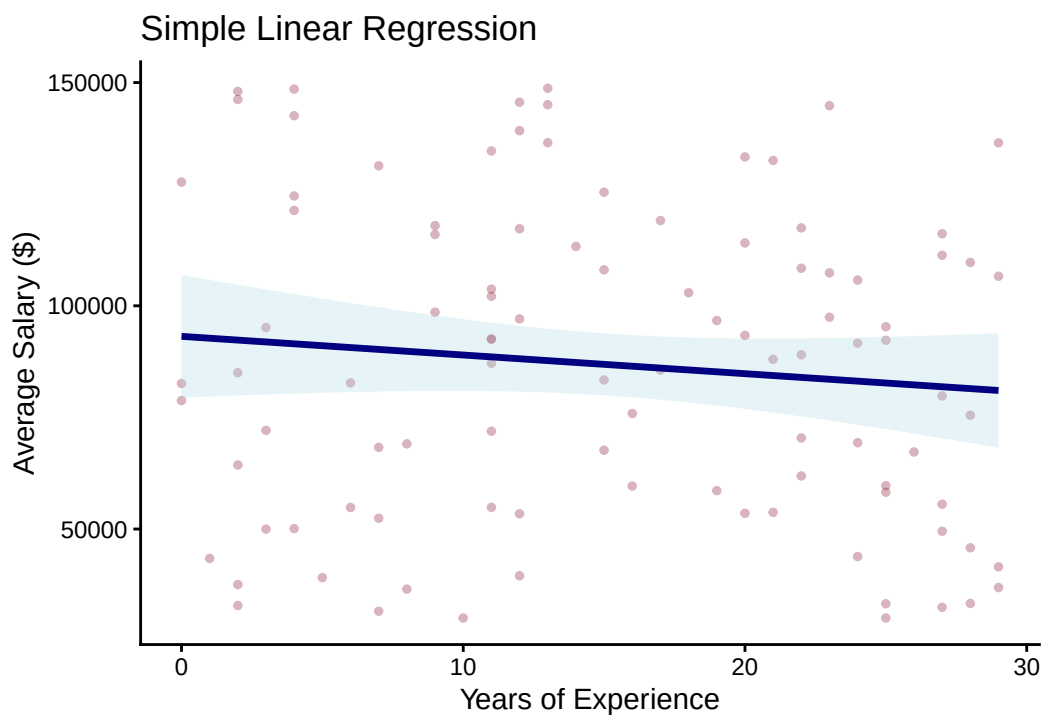


Figure 7: Regression Line: Experience vs Salary

Interpretation: The line is nearly horizontal, confirming the weak negative slope. The wide confidence

band shows high uncertainty in the predictions. The model is not useful for predicting salary from experience alone.

7.7 Predictions

```
# Create new data with experience values to predict for
# Column name must exactly match the predictor in lm()
new_data <- data.frame(Years_Experience = c(2, 10, 20, 29))

# predict() uses the fitted model to calculate predicted salaries
# interval = "confidence" gives a 95% confidence interval
preds <- predict(slr_model,
                 newdata = new_data,
                 interval = "confidence",
                 level = 0.95)

# Combine experience values with predictions
result <- data.frame(
  Experience = c(2, 10, 20, 29),
  round(preds, 2)
)
colnames(result) <- c("Experience", "Predicted",
                    "Lower CI", "Upper CI")
print(result)
```

	Experience	Predicted	Lower CI	Upper CI
1	2	92319.45	79917.23	104721.67
2	10	88975.92	80944.22	97007.61
3	20	84796.50	76909.79	92683.21
4	29	81035.03	68229.47	93840.59

Interpretation: All predicted salaries are very similar regardless of experience level. This confirms the model essentially predicts the overall mean salary for every worker, which reflects the extremely low R-squared value.

8. Conclusion

8.1 Summary of Findings

Descriptive Statistics: Average salary is \$86,807 with a standard deviation of \$34,989. AI Exposure and Automation Probability both average near 0.49. Medium risk is the most common category (49%).

Hypothesis Testing:

- t-Test: No significant salary difference between experience groups. Fail to reject H_0 .
- ANOVA: No significant salary difference across education levels. Fail to reject H_0 .
- Chi-Square: Education Level and Risk Category are independent. Fail to reject H_0 .

Correlation: All correlations with salary are very weak ($|r| < 0.10$) and not statistically significant ($p > 0.05$).

Regression: Salary = 93,155.33 - 417.94 x Experience. R-squared is only 0.0114, meaning experience explains only 1.14% of salary variation. The model is not significant overall.

8.2 Practical Implications

For workers, experience and education alone may not determine salary. Other factors like specific job skills, industry, and location likely matter more. For policymakers, since 22.9% of jobs are at high automation risk, there is a need for retraining programs to help affected workers.

8.3 Limitations

- The data is a snapshot and does not show change over time
- Automation probabilities for 2030 are estimates, not actual outcomes
- Important variables like location, industry, and specific skills are not included
- The regression model explains less than 1% of salary variation

8.4 Future Work

- Include more variables like location, industry, and skills
- Try multiple regression to improve prediction
- Study how salary actually changes from 2024 to 2030

References

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— End of Report —

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